

PJ Girls College F-8/2, Islamabad

FIRST TERM EXAMINATION

AUGUST, 2026

CLASS M-II

MATHEMATICS (SCIENCE GROUP)

Section -A (Marks 10)

Time Allowed: 15 Minutes

Section-A is compulsory. All parts of this section are to be answered on this page and handed over to the invigilator. Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER									
2	4	6	8	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

VERSION NO.			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer Sheet No. _____ Name & Section Arif - MIIH

Invigilator Sign: _____ Candidate Sign: _____



03
10

Q:1 Fill the relevant bubble against each question according to the curriculum.

Questions	A	B	C	D	A	B	C	D
1. A non-terminal, non recurring decimal represents:	A natural number	A rational number	An irrational number	A prime Number	☒	☐	☐	☐
2. Which of the following sets have the closure property w.r.t addition:	{0}	{0, -1}	{0, 1}	$\{1, \sqrt{2}, \frac{1}{2}\}$	☐	☐	☒	☒
3. Base of natural logarithm is:	Rational number	Integer	Irrational Number	10	☐	☒	☐	☐
4. $\log_{\sqrt{10}} 100^2$ is:	2	1	4	8	☒	☐	☐	☐
5. What should be added in $4a^2 - 24ab$ to complete the square?	$6b^2$	$36b^2$	$3b^2$	$4b^2$	☐	☒	☐	☐
6. A point p lies on x - axis with vertices:	(2,3)	(0,4)	(5,0)	(-2,3)	☐	☒	☐	☐
7. Which is the solution set of $ x + 7 = 3$	{-4}	{4, -4}	{}	{-7, -3}	☐	☒	☐	☐
8. Which of the following is divided by a point called mid point?	An angle	A line	A line segment	A ray	☐	☐	☒	☐
9. Minimum number of sides of a polygon is:	2	3	4	5	☐	☒	☐	☐
10. What is the least common multiple of $12x^2y^2, 6xy^3, 24x^2y^2$	$-6xy$	$6x^2y^2$	$6xy^2$	$+24x^2y^2$	☐	☐	☒	☐

O.P.F. Girls College Islamabad

Name Ademab Afif

Roll # 24683

Class / Section m-2 H

Subject Mathematics

Examination First term exams

Date 4-Sep-2026



Q. #	1	2	3	4	5	7	8	9	10	11	12	13	14	Total
Marks Obtained	03/23	10/24	8/8	4/8										38/56

Invigilator [Signature]

Examiner [Signature]

Rechecked by _____

Q: 2(i)(or)

Ans: 2(i) (or)

$$\frac{(216)^{\frac{2}{3}} \times (25)^{\frac{1}{2}}}{(0.04)^{-\frac{3}{2}}}$$

$$\frac{(6^3)^{\frac{2}{3}} \times (5^2)^{\frac{1}{2}}}{(0.04)^{-\frac{3}{2}}}$$

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$$\left(\frac{6^2 \times 5}{(0.04)^{-\frac{3}{2}}} \right)^{\frac{1}{2}}$$

$$\left(\frac{6^2 \times 5}{(0.2)^{-3}} \right)^{\frac{1}{2}}$$

$$\left(\frac{6^2 \times 5}{0.2^{-3}} \right)^{\frac{1}{2}}$$

$$\frac{6^{\frac{2}{2}} \times 5^{\frac{1}{2}}}{0.2^{\frac{3}{2}}}$$

$$\frac{6 \times 5^{\frac{1}{2}}}{(0.2)^{-\frac{3}{2}}}$$

?

Q.2(ii)(or)

A: 2(ii)(or)

a) $4x^2 - y^2 - 2y - 1$

$4x^2 - y^2 - 2y - 1$

$4x^2 - (y^2 + y + y + 1)$

$4x^2 - (y(y+1) + y(y+1))$

$4x^2 - (y^2 - y)(y+1)$

$4x^2 - (y-1)^2$

$(2x)^2 - (y-1)^2$

$(2x - y + 1)(2x - y - 1)$

4/4

$4x^2 - (y^2 + 2(y)(1) + (1)^2)$

$4x^2 - (y+1)^2$

4/4

$(2x)^2 - (y+1)^2$

$(2x - y - 1)(2x - y + 1)$

Q.2(iii)

A: 2(iii)

4/4

$ax^2 + 2ax + a^3, 2ax^2 - 4ax - 6a^3, 3(ax + a^2)^2$

$a(x^2 + 2ax + a^2), 2a(x^2 - 2ax - 3a^2), 3a(x+a)^2$

$a(x^2 + 2ax + a^2), 2a(x^2 - 3ax - 3a^2), 3a(x+a)^2$

$a(x(x+a) + a(x+a)), 2a(x(x-3a) + a(x-3a)), 3a(x+a)(x+a)$

$a(x+a)(x+a), 2a(x+a)(x-3a), 3a(x+a)(x+a)$

HCF :- $(x+a) \cdot a$

Q.2(iv)

Given :-

A(-1, 2)

B(7, 5)

C(-2, 6)

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(7 - (-1))^2 + (5 - 2)^2}$$

$$= \sqrt{(7+1)^2 + (3)^2}$$

$$= \sqrt{(8)^2 + 9}$$

$$= \sqrt{64 + 9}$$

$$= \sqrt{73}$$

4/4

$$d_{BC} = \sqrt{(-2-7)^2 + (6-5)^2}$$

$$= \sqrt{(-9)^2 + (1)^2}$$

$$= \sqrt{81 + 1}$$

$$= \sqrt{82}$$



$$d_{AC} = \sqrt{(-2 - (-1))^2 + (6 - 2)^2}$$

$$= \sqrt{(-2+1)^2 + (4)^2}$$

$$= \sqrt{(-1)^2 + 16}$$

$$= \sqrt{1 + 16}$$

$$= \sqrt{17}$$

Pythagoras theorem:-

$$AB^2 + BC^2 = AC^2$$

$$AB^2 + AC^2 = BC^2$$

$$\sqrt{73}^2 + \sqrt{17}^2 = \sqrt{82}^2$$

$$73 + 17 = 82$$

$$90 \neq 82$$

Not right angle triangle

$$Q: 2(v)(or)$$

$$A: 2(v)(or)$$

$$\frac{x^3-8}{x^2-4} \times \frac{x^2+6x+8}{x^2-2x+1}$$

4/4

$$\frac{x^3-(2)^3}{(x^2)-(2)^2} \times \frac{x^2+4x+2x+8}{(x)^2-2(x)(1)+(1)^2}$$

$$\frac{(x-2)(x^2+2x+2^2)}{(x-2)(x+2)} \times \frac{x(x+4)+2(x+4)}{(x-1)}$$

$$\frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} \times \frac{(x+2)(x+4)}{(x-1)(x+1)}$$

$$\frac{x^2+2x+4}{(x-1)(x+1)} \times \frac{x+4}{(x-1)(x+1)}$$

$$Q: 2(v)$$

$$\sqrt{\frac{2}{2+3}} = \sqrt{\frac{2+2}{2+6}}$$

4/4

Taking squares on b.s :-

$$\left(\sqrt{\frac{2}{2+3}}\right)^2 = \left(\sqrt{\frac{2+2}{2+6}}\right)^2$$

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Signature Of Invigilator

Copy No:

Name: Roll No:

Class: Section: Subject:

Date: No Of Continuation Sheets Attached:

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

Start writing here

$$\frac{z}{z+3} = \frac{z+2}{z+6}$$

{ multiplication cross }

$$z(z+6) = z+2(z+3)$$

$$z^2 + 6z = z + 2(z+3)$$

$$z^2 + 6z = z^2 + 3z + 2z + 6$$

$$z^2 + 6z = z^2 + 5z + 6$$

$$z^2 - z^2 + 6z = -5z + 6$$

$$+6z = -5z + 6$$

$$z = 6$$

Check :-

$$\sqrt{\frac{6}{6+3}} = \sqrt{\frac{6+2}{6+6}}$$

$$\sqrt{\frac{6^2}{9 \cdot 3}} = \sqrt{\frac{8^2}{12 \cdot 3}}$$

$$\sqrt{\frac{2}{3}} = \sqrt{\frac{2}{3}}$$

$$\{ 6 \}$$

Section - C

Q.3

A : 3

$$\frac{37.54 \times (38.4)^{\frac{1}{2}}}{\sqrt[3]{31.5}}$$

Taking $\log$:-

$\frac{8}{8}$

$$\log u = \log(37.54)^{\frac{1}{2}} \times (38.4)^{\frac{1}{2}} - \log(31.5)^{\frac{1}{3}}$$

$$\log u = \log 37.54^{\frac{1}{2}} + \log 38.4^{\frac{1}{2}} - \log 31.5^{\frac{1}{3}}$$

$$= \frac{1}{2} \log 37.54 + \frac{1}{2} \log 38.4 - \frac{1}{3} \log 31.5$$

$$= \frac{1}{2} (1.5744) + \frac{1}{2} (1.5843) - \frac{1}{3} (1.4983)$$

$$= 0.7872 + 0.79215 - 0.4994$$

$$= 1.08035$$

Taking antilog:-

$$\text{Antilog}_{1.08035} = 12.032$$

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Signature Of Invigilator

[Signature]

Copy Not

Name: Seemab Atif

Roll No. 24683

Class: M-2 H

Section: H

Subject: Mathematics

Date: 4-9-2026

No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

Start writing here

Q:4

$$\cancel{x^3} - 6x^2$$

A:4

$$(x^3 - 6x^2y + 12xy^2 - 8y^3)m^3$$

04/8

① Expression for length of each side :-

$$\begin{aligned} & (x^3 - 6x^2y + 12xy^2 - 8y^3)m^3 \\ 1) & (x^3 - 3(x)(2) + 3(x)(2)^2 - (2y)^3) \\ & (x - 2y)^3 m^3 \\ & L = ? \end{aligned}$$

② Expression for area of base :- L^3 ✓

$$\cancel{(x - 2y)^3 m^3}$$

③ Expression for surface area :- $6L^2$ ✓

$$\begin{aligned} 2) & 6(x - 2y)^2 \\ & (6x - 12y) \end{aligned}$$

④ Cost :-

$$\begin{aligned} 1) & 50(6x - 12y) \\ & 300x - 600y \end{aligned}$$